

Programme	B. Sc. Mathematics Honours			
Course Code	MAT2MN101			
Course Title	DIFFERENTIAL EQUATIONS AND MATRIX THEORY			
Type of Course	Minor			
Semester	II			
Academic Level	100 –199			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Basic Calculus			
Course Summary	This course covers a range of topics. It starts with introducing fundamental terminology and methods for solving differential equations, including separable equations, linear equations, exact equations, and equations with constant coefficients. Then it proceeds into more specialized topics such as homogeneous linear equations with constant coefficients and Cauchy-Euler equations, providing methods for their solution. Laplace transforms, including their definition, properties, and applications in solving differential equations and transforming derivatives are explored. The course concludes with an introduction to vector spaces, matrix theory and the eigenvalue problem, Fourier series, and separable partial differential equations, providing a comprehensive foundation in advanced calculus and its applications to engineering and physics.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Solve basic ordinary differential equations using separation of variables, linear methods, and Laplace transforms.	Ap	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO2	Apply concepts from linear algebra, including matrices, determinants, and eigenvalues, to solve systems of equations and analyse linear systems.	Ap	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO3	Analyse periodic functions using Fourier series and solve separable partial differential equation	An	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text		Advanced Engineering Mathematics, 6 th Edition, Dennis G. Zill, Jones & Bartlett Learning LLC (2018) ISBN: 978-1-284-10590-2		
Module		Content	Hrs (48 +12)	Ext. Marks (70)
		Differential Equations		
I	1	Introduction to Differential Equations - Section 1.1: Definitions and Terminology - A Definition, Classification by Type, Notation, Classification by Order, Classification by Linearity, Solution (with examples)	11	Min 15
	2	Section 2.2: Separable Equations - Introduction, A Definition, Method of Solution (with examples)		
	3	Section 2.3: Linear Equations - Introduction, A Definition, Standard Form, Method of Solution, An Initial Value Problem (Examples 4 & 5, ref section 1.1)		
	4	Section 2.4: Exact Equations - Introduction, Differential of a Function of Two Variables, Method of Solution.		
	5	Section 3.3: Homogeneous Linear Equations with Constant Coefficients - Introduction, Auxiliary Equation.		
	6	Section 3.6: Cauchy-Euler Equations - Cauchy-Euler Equation (Second Order Only), Method of Solution.		
		Laplace Transforms		
II	7	Section 4.1: Definition of the Laplace Transform - Basic Definition (Definition 4.1.1 onwards)	14	Min 15
	8	Section 4.1: Definition of the Laplace Transform - L is a Linear Transform.		
	9	Section 4.2: The Inverse Transform and Transforms of Derivatives - Inverse Transforms		
	10	Section 4.2: The Inverse Transform and Transforms of Derivatives - Transforms of Derivatives		
	11	Section 7.6: Vector Spaces - Vector Space (Example 2 is optional), Subspace.		
	12	Section 7.6: Vector Spaces - Basis, Standard Bases, Dimension, Span		
		Matrix Theory	13	Min 15
III	13	Section 8.2: Systems of Linear Algebraic Equations - Introduction, General Form, Solution, Augmented Matrix, Elementary Row Operations, Elimination Methods.		
	14	Section 8.2: Systems of Linear Algebraic Equations - Homogeneous Systems, Notation		
	15	Section 8.3: Rank of a Matrix - Introduction, A Definition, Row Space, Rank by Row Reduction, Rank and Linear Systems.		

	16	Section 8.4: Determinants - Introduction, A Definition (Topics up to and including Example 2).		
	17	Section 8.8: The Eigenvalue Problem - Introduction, A Definition (Topics up to and Including Example 4)		
	18	Section 8.8: The Eigenvalue Problem - Eigenvalues and Eigenvectors of A^{-1} .		
IV	Fourier Series and PDE			
	19	Section 12.2: Fourier Series - Trigonometric Series (Definition 12.2.1 onwards), Convergence of a Fourier Series.	10	Min 15
	20	Section 12.3: Fourier Cosine and Sine Series - Introduction, Even and Odd Functions, Properties, Cosine and Sine Series (Definition 12.3.1 onwards).		
	21	Section 13.1: Separable Partial Differential Equations - Introduction, Linear Partial Differential Equation, Solution of a PDE, Separation of Variables.		
	22	Section 13.1: Separable Partial Differential Equations - Classification of Equations.		
	Open Ended			
	1	Initial-Value Problems	12	
	2	Method of Integrating Factors		
	3	Differential Equations as Mathematical Models		
	4	Second Order Non-Homogeneous Equations-Method of Undetermined Coefficients, Variation of Parameters.		
	5	Linear Models – IVP and their solutions by Laplace Transform		
	6	Linear Models - BVP		
	7	Non-linear Models		
	8	Complex Eigen Values		
	9	Half- Range Fourier Series		
	10	Classical PDEs and Boundary- Value Problems		
	1	Advanced Engineering Mathematics, Erwin Kreyszig, 10 th Edition, Wiley India.		
	2	Calculus & Analytic Geometry, 9 th Edition, George B. Thomas & Ross L. Finney, Pearson Publications.		
	3	Calculus, 7 th Edition, Howard Anton, Biven, & Stephen Davis, Wiley India.		

Note: 1) Optional topics are exempted for end semester examination.

2) Proofs of all the results are also exempted for the end semester exam.

Mapping of COs with PSOs and POs :

	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	1	3	1	3	3	3	1	2
CO 2	2	1	3	1	3	3	3	1	2
CO 3	3	2	3	2	3	3	3	2	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓